

TASK 1:

```

Last login: Wed Aug 12 14:45:18 on tty987
amruthasekuthus-MacBook-Air: ~ % brew install postgresql@17
Warning: postgresql@17 17.10.1 is already installed and up-to-date.
To reinstall 17.10.1, run:
  brew reinstall postgresql@17
amruthasekuthus-MacBook-Air: ~ % brew services start postgresql@17
Service "postgresql@17" already started, use 'brew services restart postgresql@17' to restart.
amruthasekuthus-MacBook-Air: ~ % psql --version
psql (PostgreSQL) 17.10 (Homebrew)
amruthasekuthus-MacBook-Air: ~ % psql postgres
psql (17.10 (Homebrew))
Type "help" for help.

postgres=# CREATE DATABASE bookflow_db;
ERROR: database "bookflow_db" already exists
postgres=# \c bookflow_db
You are now connected to database "bookflow_db" as user "amrutha".
bookflow_db=# CREATE EXTENSION IF NOT EXISTS vector;
NOTICE: extension "vector" already exists, skipping
CREATE EXTENSION
bookflow_db=# \dt
Did not find any relations.
bookflow_db=# CREATE TABLE books (
  book_id SERIAL PRIMARY KEY,
  title VARCHAR(100),
  isbn VARCHAR(30),
  published_year INT
);
CREATE TABLE
bookflow_db=# INSERT INTO books (title, isbn, published_year) VALUES
('Python Programming', 'ISBN001', 2022),
('Database Systems', 'ISBN002', 2021),
('Machine Learning', 'ISBN003', 2024);
INSERT 0 3
bookflow_db=# SELECT * FROM books;
 book_id | title           | isbn      | published_year
-----+-----+-----+-----
      1 | Python Programming | ISBN001   |          2022
      2 | Database Systems   | ISBN002   |          2021
      3 | Machine Learning   | ISBN003   |          2024
(3 rows)

bookflow_db=# ALTER TABLE books
ADD COLUMN embedding VECTOR(3);
ALTER TABLE
bookflow_db=# \d books
          Column          | Type          | Table "public.books"
          |              | Collation | Nullable | Default
-----+-----+-----+-----+-----
 book_id | integer       |           | not null | nextval('books_book_id_seq'::regclass)
  title  | character varying(100) |           |          |
  isbn   | character varying(30)  |           |          |
 published_year | integer       |           |          |
 embedding | vector(3)       |           |          |
Indexes:
  "books_pkey" PRIMARY KEY, btree (book_id)

bookflow_db=#

```

TASK 2:

```

ERROR: syntax error at or near "["
LINE 1: [200-UPDATE books
^
bookflow_db=# UPDATE books
SET embedding = '[0.20,0.80,0.10]'
WHERE book_id = 2;
ERROR: syntax error at or near "-"
LINE 1: -
^
bookflow_db=# \r
Query buffer reset (cleared).
bookflow_db=# UPDATE books
SET embedding = '[0.20,0.80,0.10]'
WHERE book_id = 2;
UPDATE 1
bookflow_db=# SELECT book_id, title, embedding
FROM books;
 book_id | title           | embedding
-----+-----+-----
      1 | Python Programming |
      3 | Machine Learning   |
      2 | Database Systems   | [0.2,0.8,0.1]
(3 rows)

bookflow_db=# SELECT book_id, title, embedding
FROM books;
 book_id | title           | embedding
-----+-----+-----
      1 | Python Programming |
      3 | Machine Learning   |
      2 | Database Systems   | [0.2,0.8,0.1]
(3 rows)

bookflow_db=# SELECT
  book_id,
  title,
  embedding <-> '[0.85,0.15,0.60]' AS distance
FROM books
WHERE embedding IS NOT NULL
ORDER BY embedding <-> '[0.85,0.15,0.60]'
LIMIT 3;
 book_id | title           | distance
-----+-----+-----
      2 | Database Systems | 1.046422490493314
(1 row)

bookflow_db=# SELECT
  book_id,
  title,
  embedding <=> '[0.85,0.15,0.60]' AS cosine_distance
FROM books
WHERE embedding IS NOT NULL
ORDER BY embedding <=> '[0.85,0.15,0.60]'
LIMIT 3;
 book_id | title           | cosine_distance
-----+-----+-----
      2 | Database Systems | 0.5991679991952114
(1 row)

bookflow_db=#

```

TASK 3:

```
amrutha@Amruthas-MacBook-Air ~ % mkdir -p ~/bookflow_api
amrutha@Amruthas-MacBook-Air ~ % cd ~/bookflow_api
amrutha@Amruthas-MacBook-Air bookflow_api % nano openapi.yaml
amrutha@Amruthas-MacBook-Air bookflow_api % ls
openapi.yaml
amrutha@Amruthas-MacBook-Air bookflow_api % cat openapi.yaml
openapi: 3.0.3
```

```
info:
  title: BookFlow Semantic Search API
  version: 1.0.0
  description: API for semantic book search

paths:
  /v1/books/search:
    get:
      summary: Search books semantically

      parameters:
        - name: q
          in: query
          required: true
          schema:
            type: string
            description: Search query

        - name: limit
          in: query
          required: false
          schema:
            type: integer
            minimum: 1
            maximum: 50
            default: 10
            description: Maximum number of results

      responses:
        '200':
          description: Search results
          content:
            application/json:
              schema:
                type: object
                properties:
                  results:
                    type: array
                    items:
                      type: object
                      properties:
                        book_id:
                          type: integer
                        title:
                          type: string
                        distance:
                          type: number

        '400':
          description: Bad Request
          content:
            application/json:
```

```
type: object
properties:
  results:
    type: array
    items:
      type: object
      properties:
        book_id:
          type: integer
        title:
          type: string
        distance:
          type: number
```

```
'400':
  description: Bad Request
  content:
    application/json:
      schema:
        type: object
        properties:
          error:
            type: object
            properties:
              code:
                type: string
              message:
                type: string
```

TASK 4:

```
amrutha@Amruthas-MacBook-Air bookflow_api % cat openapi.yaml
openapi: 3.0.3
```

```
info:
  title: BookFlow Semantic Search API
  version: 1.0.0
  description: API for semantic book search
```

```
paths:
  /v1/books/search:
    get:
      summary: Search books semantically

      parameters:
        - name: q
          in: query
          required: true
          schema:
            type: string
            description: Search query

        - name: limit
          in: query
          required: false
          schema:
            type: integer
            minimum: 1
            maximum: 50
            default: 10
            description: Maximum number of results
```

```
responses:
  '200':
    description: Search results
    content:
      application/json:
        schema:
          type: object
          properties:
            results:
              type: array
              items:
                type: object
                properties:
                  book_id:
                    type: integer
                  title:
                    type: string
                  distance:
                    type: number
```

```
  '400':
    description: Bad Request
    content:
      application/json:
        schema:
          type: object
          properties:
            error:
              type: object
```

```
        results:
          type: array
          items:
            type: object
            properties:
              book_id:
                type: integer
              title:
                type: string
              distance:
                type: number

  '400':
    description: Bad Request
    content:
      application/json:
        schema:
          type: object
          properties:
            error:
              type: object
              properties:
                code:
                  type: string
                message:
                  type: string
```

```
amrutha@Amruthas-MacBook-Air bookflow_api % curl -X GET "http://localhost:3000/v1/books/search?q=stories%20about%20space&limit=3"
curl: (7) Failed to connect to localhost port 3000 after 6 ms: Couldn't connect to server
amrutha@Amruthas-MacBook-Air bookflow_api % GET /v1/books/search?q=stories%20about%20space&limit=3
[1] 27818
zsh: no matches found: /v1/books/search?q=stories%20about%20space
amrutha@Amruthas-MacBook-Air bookflow_api %
[1] + exit 1 GET /v1/books/search?q=stories%20about%20space
amrutha@Amruthas-MacBook-Air bookflow_api % grep -A 20 "/v1/books/search" openapi.yaml
```

```
/v1/books/search:
  get:
    summary: Search books semantically

    parameters:
      - name: q
        in: query
        required: true
        schema:
          type: string
          description: Search query

      - name: limit
        in: query
        required: false
        schema:
          type: integer
          minimum: 1
          maximum: 50
          default: 10
          description: Maximum number of results
```

```
amrutha@Amruthas-MacBook-Air bookflow_api % grep -E "'200'|'400'" openapi.yaml
  '200':
  '400':
amrutha@Amruthas-MacBook-Air bookflow_api %
```